LAB 1: AI-ASSISTED CITATION INTEGRITY AUDIT

Generate a Research Paper with AI, Then Audit Its Citations

AI Tools for Research • 2 Hours • Individual Activity • Topic IDs T1-T40

Student Detail	Fill in
Name	SRISHTI KUMARI
Roll Number	RSI2026514
Programme / Department	IT
Topic ID Full Assigned Topic	29 Comparing Explainability Outputs of AI Copilots and Manual SHAP/LIME Analysis
AI Tool Used	Gemini
Date	21/08/2026

IMPORTANT: You will receive a topic from T1-T40. Your task is not to produce the “best” paper. Your task is to generate an AI-written research paper and systematically determine whether its citations can be trusted.

1. PURPOSE OF THIS LAB

Generative AI systems can produce research papers that look scholarly and may contain convincing references, author names, journal titles, DOIs, arXiv IDs, and in-text citations. However, the presence of a citation does not guarantee that the cited publication actually exists, that its metadata are correct, that the identifier belongs to that publication, or that the cited paper supports the statement made by the AI.

In this laboratory activity, you will test these issues yourself by generating a research paper and auditing a systematic sample of its references.

Question 1 - Citation Authenticity: Does the cited publication actually exist, and is it correctly identified? Question 2 -

Claim-Citation Support: If the publication is genuine, does it actually support the claim for which the AI cited it?

2. IMPORTANT RULES

● Use only the topic assigned to you by the instructor.
● Generate the paper in a fresh AI conversation.
● Do not tell the AI that you are conducting a citation audit or hallucination test.
● Do not intentionally ask the AI to generate fake references.
● Preserve the original AI-generated paper exactly as produced.
● Do not correct, regenerate, replace, or question references before saving the original output. ● For verification, do not accept another AI chatbot's statement as proof that a reference exists. ● Use scholarly databases, publisher pages, DOI/arXiv/PMID records, or other reliable scholarly sources. ● Record what you actually find, even if all references are genuine or all are problematic.

Most important principle: You are not being evaluated on whether the AI produces good or bad references. You are being evaluated on how systematically and correctly you audit them.

3. WHAT YOU WILL DO

Part	Activity
A	Record your baseline perception of AI citations
B	Record the AI environment used
C	Generate and preserve an AI-written research paper
D	Create a reference inventory
E	Select 10 references systematically
F G	Judge their plausibility before verification Verify authenticity, metadata and identifiers
H	Determine whether genuine references support the AI-generated claims

4. RECOMMENDED 2-HOUR TIME PLAN

Time	Activity
0-10 min	Parts A-B: Baseline and AI environment
10-30 min	Part C: Generate and save research paper
30-40 min	Parts D-E: Inventory and select references
40-45 min	Part F: Pre-verification plausibility
45-80 min	Part G: Authenticity and metadata audit
80-110 min	Part H: Claim-citation entailment audit
110-120 min	Calculate scores, check worksheet and prepare submission

Do not spend excessive time formatting the AI-generated paper. The audit is the main activity.

PART A - BASELINE ASSESSMENT

Complete this section BEFORE opening the AI tool.

A1. How often do you use generative AI for academic work?

☐ Never ☐ Rarely ☐ Occasionally ☐ Weekly ☒ Almost daily

A2. Have you previously asked an AI system to generate references?

☐ Frequently ☐ Occasionally ☒ Once or twice ☐ Never

A3. Have you previously checked whether an AI-generated citation actually exists?

☐ Always ☐ Sometimes ☐ Rarely ■ Never ☐ Not applicable

A4. Before this lab, how confident are you that citations generated by AI are generally reliable?

☐ 1 ☐ 2 ☐ 3 ■ 4 ☐ 5 (1 = not at all confident; 5 = completely confident)

Important: Do not change this answer later, even if your opinion changes during the experiment.

PART B - RECORD YOUR AI ENVIRONMENT

Information	Your Response
AI tool used	Gemini
Model name	Gemini 3.6 flash
Model/version shown	Not shown
Access tier Web browsing available?	Free Yes
Was web browsing actually used? Research/Deep Research mode used? Date/time generated	No ☐ Yes ☐ No 21/08/2026 at 10:00 am

Important: If the model/version is not displayed, write NOT SHOWN. Do not guess the model version.

PART C - GENERATE THE AI RESEARCH PAPER

C1. Open a Fresh AI Conversation

Do not tell the AI that this is a citation audit, that you are testing hallucinations, or that you expect fabricated references. We want to observe normal AI research-paper generation behaviour.

C2. Use the Following Common Prompt

Replace only [YOUR ASSIGNED TOPIC].

> Generate a scholarly research paper on the topic: “[YOUR ASSIGNED TOPIC]”. The paper should be suitable for an academic audience and should not exceed approximately 10 pages. Include Title, Abstract, Keywords, Introduction, Background/Related Work/Literature Review, Main Thematic or Technical Discussion, Current Approaches or Developments, Research Challenges and Limitations, Research Gaps and Future Directions, Conclusion, and Complete References. Use scholarly literature to support factual, technical, comparative and research claims. Provide in-text citations throughout the paper. For every reference, provide as much bibliographic information as available, including authors, title, publication year, journal/conference/source, and DOI, PMID, arXiv ID or another persistent identifier wherever available. Use genuine scholarly sources. Do not intentionally fabricate references.

C3. Preserve the Original Output

☐ Save the complete AI-generated paper

☐ Save the complete reference list

☐ Take screenshot(s) showing the generated output/references

☐ Save conversation/export/share evidence if available

STOP: Do not ask the AI to correct or verify anything yet. Your original output is the experimental evidence.

PART D - PAPER PROFILE AND REFERENCE INVENTORY

Paper title: Comparing Explainability Outputs of AI Copilots and Manual SHAP/LIME Analysis.

Measure	Value
Approximate pages Approximate word count	5 1350
Number of sections	7
Total number of references Approximate in-text citation occurrences Did AI warn you to verify references?	10 10 in-text citation instances throughout the text body. No

D1. Reference Inventory

Classify every reference once using the best available identifier in this priority order: DOI -> PMID -> arXiv ID -> URL only -> No persistent identifier. The category totals must equal the total number of references in the AI-generated paper.

Reference Type	Count
DOI available	6
PMID available, no DOI	0
arXiv ID available, no DOI/PMID	2
URL only	2
No persistent identifier supplied	0
TOTAL	10

Examples

Category	Example	How to Count
DOI available	Kather, J. N., et al. (2019). Predicting survival from colorectal cancer histology slides using deep learning. PLoS Medicine. DOI: 10.1371/journal.pmed.1002730	DOI available
PMID available, no DOI	A biomedical paper for which the AI provides PMID: 12345678, but no DOI is supplied/found	PMID available, no DOI
arXiv ID available	Yao, S., et al. ReAct: Synergizing Reasoning and Acting in Language Models.	arXiv ID available

	arXiv:2210.03629	
URL only	World Health Organization. Colorectal cancer. Available at: https://... with no DOI, PMID or arXiv ID	URL only
No persistent identifier	Goodfellow, I., Bengio, Y., & Courville, A. (2016). Deep Learning. MIT Press.	No persistent identifier if none is supplied

Important: Absence of a DOI does not mean a reference is fake. An identifier being present also does not prove authenticity; it must be checked.

PART E - SELECT 10 REFERENCES FOR AUDIT

You are not allowed to choose only suspicious-looking references. This prevents selection bias.

If the bibliography contains 10 or fewer references: audit all references.

If it contains more than 10 references: select the first 3 references, the last 3 references, and 4 references approximately evenly distributed through the middle.

Example: If there are 28 references, a reasonable sample could be 1, 2, 3, 8, 13, 18, 23, 26, 27, 28.

Audit Slot	Original Bibliography Reference No.
R1	1
R2	2
R3	3
R4	4
R5	5
R6	6
R7	7
R8	8
R9	9
R10	10

Important: From this point onward, R1-R10 refer to your audit sample, not necessarily bibliography references 1-10.

PART F - PLAUSIBILITY TEST

Do this BEFORE searching for any reference.

Look only at the citation as generated by the AI. Ask yourself: “If I had not verified this citation, how convincing would it look to me?”

Score	Meaning
1	Obviously suspicious

2	Somewhat suspicious
3	Unsure
4	Looks genuine
5	Completely convincing

Ref.	Initial Impression	Plausibility 1-5	Predict Genuine?
R1	Standard, highly detailed survey citation with well-formed DOI .	5	Yes
R2	standard AAAI conference formatting, and valid-looking DOI string.	5	Yes
R3	Canonical seminal paper format	5	Yes
R4	direct ACM DL link format	5	Yes
R5	standard KDD conference entry with official DOI.	5	Yes
R6	Standard follow-up paper	5	Yes
R7	Well-known	5	Yes
R8	Recent high-impact NeurIPS paper	5	Yes
R9	Standard legal/tech journal citation	5	Yes
R10	Have complete author list and valid arXiv ID format.	5	Yes

Example: A citation with realistic authors, a well-known journal, complete volume/pages and a DOI-like identifier may look highly convincing. You might record Plausibility = 5 and Predict Genuine = Yes. You are recording your pre-verification belief, not the truth.

Critical rule: Never change Part F after completing Part G. Your incorrect predictions are valuable experimental observations.

PART G - CITATION AUTHENTICITY AND METADATA AUDIT

G1. What Are You Checking?

- Does the publication exist?
- Is the title correct?
- Are the authors correct?

- Is the year correct?
- Is the journal/conference/source correct?
- Are volume, issue and pages correct, if supplied?
- Does the DOI/arXiv/PMID belong to this publication?

G2. Verification Procedure

☐ Persistent identifier: DOI / arXiv / PMID

☐ Publisher or official repository

■ Google Scholar

☐ Semantic Scholar / OpenAlex

☐ Crossref

☐ PubMed or another discipline-specific database

☐ Exact-title search

☐ Author + title-keyword search

Not acceptable as sole evidence: “I asked another AI and it said the paper exists.” AI can help you search, but final verification must rely on scholarly or publisher evidence.

G3. Assign One A-E Code

Code	Classification	Meaning
A	VERIFIED	Publication exists and important metadata arecorrect.
B	WRONG METADATA	Publication exists, but one or more bibliographic details are incorrect.
C	FRANKENSTEIN	Real-looking pieces are combined, but that specific publication does not exist.
D	FABRICATED	No reliable scholarly evidence that the publicationexists.
E	IDENTIFIER MISMATCH	DOI/arXiv/PMID is invalid or resolves to a different publication.

G4. Examples

A: AI gives a paper and all details match the publisher record.

B: The paper exists, but the AI gives the wrong publication year or journal.

C: The authors are real, the journal is real, and the topic is plausible, but no such paper with that title/authorship combination exists.

D: You cannot find the paper or a credible scholarly record after systematic searching.E: The DOI supplied by AI opens an entirely different paper or does not resolve.

G5. Record Your Evidence

Ref.	Publication Found?	Title Match?	Authors Match?	Year Match?	Venue Match?	Identifier Match?

R1	Y	Y	Y	Y	Y	Y
R2	Y	Y	Y	Y	Y	Y
R3	Y	Y	Y	Y	Y	Y
R4S	Y	Y	Y	Y	Y	Y
R5	Y	Y	Y	Y	Y	Y
R6	Y	Y	Y	Y	Y	Y
R7	Y	Y	Y	Y	Y	Y
R8	Y	Y	Y	Y	Y	Y
R9	Y	Y	Y	Y	Y	Y
R10	Y	Y	Y	Y	Y	Y

Ref.	Verification Source(s)	Identifier Checked	What You Found
R1	ACM Digital Library / Crossref	DOI: 10.1145/3236009	Official record confirms exact title, authors, issue 5, and page numbers 1–42.
R2	AAAI Publications	DOI: 10.1609/hccs.v8i1.138 28	Exact match for title, authors (Lakkaraju et al.), AAAI HCOMP venue, and pages 131–138.
R3	ACM Digital Library	DOI: 10.1145/3236386.32413 40	Exact match for Lipton (2018) in ACM Queue, Vol 16, Issue 3
R4	NeurIPS Proceedings / ACM DL	URL: dl.acm.org/doi/10.555 5/3295222.3295230	Official NeurIPS 2017 paper entry; exact page numbers 4765–4774

R5	ACM SIGKDD / Crossref	DOI: 10.1145/2939672.29397 78	Found exact match for Ribeiro et al. LIME paper, 2016 SIGKDD, pages 1135–1144.
R6	AAAI Proceedings	DOI: 10.1609/aaai.v32i1.11 491	Verified official record for Anchors paper, AAAI 2018, pages 1527–1535.
R7	PMLR Proceedings	URL: proceedings.mlr.press /v70/sundararajan17a. html	Official ICML 2017 proceedings record matching integrated gradients title and authors.
R8	NeurIPS Proceedings / arXiv	arXiv: 2305.04388	Exact match on arXiv and NeurIPS 2023 for

			Turpin et al. CoT faithfulness paper
R9	Harvard JOLT / SSRN	DOI: 10.2139/ssrn.3063289	Official SSRN repository record matching Wachter et al. counterfactual explanations

R10 NeurIPS Proceedings / arXiv arXiv: 2201.11903

Verified match on arXiv and NeurIPS 2022 for Wei et al. Chain-of-Thought paper

G6. Final Classification

Ref.	Final Code	One-Line Reason
R1	A	Publication exists and all bibliographic details including DOI match perfectly.
R2	A	Article exists and all metadata including authors, volume, and DOI are correct.
R3	A	Paper exists with exact matching metadata in ACM Queue.
R4	A	Verified publication in NeurIPS 2017 with complete matching metadata.
R5	A	Landmark KDD 2016 paper verified with matching title, authors, and DOI.
R6	A	Verified AAAI 2018 publication with exact match on DOI and pagination
R7	A	Verified ICML 2017 PMLR publication with exact author and title match.
R8	A	verified NeurIPS 2023 paper with exact matchingarXiv ID and authors.
R9	A	Verified Harvard JOLT/SSRN paper with accurate metadata and DOI link
R10	A	Seminal NeurIPS 2022 paper verified with correctarXiv ID and metadata.

Good reason: B: Article exists, but AI reported 2022 instead of 2020.

Poor reason: B: Details wrong.

PART G - SCORING

Count your final classifications: A = 10 B = 0 C = 0 D = 0 E = 0

Total audited: N = A + B + C + D + E = 10

Code	Points

A	4

B	3
C	1
D	0
E	1

Authenticity Points = 4A + 3B + C + E

Authenticity Score = [(4A + 3B + C + E) / (4N)] x 100

My calculation: 4(10) + 3(0) + 1(0) + 0(0) + 1(0) = 40 points

Authenticity Score = 40 / (4 x 10) x 100 = 100

Worked example: If A=6, B=2, C=1, D=1, E=0, then points = 4(6)+3(2)+1(1)=31; maximum = 40; Authenticity Score = 31/40 x 100 = 77.5.

Score	Interpretation
90-100	Excellent authenticity
75-89	Good, minor concerns
60-74	Moderate citation-authenticity risk
40-59	High citation-authenticity risk
Below 40	Very high citation-authenticity risk

G7. Prediction Accuracy

Compare your Part F Genuine Yes/No predictions with the Part G results.

Prediction Accuracy = (Correct genuine/fake predictions / Total predictions) x 100

Correct predictions = 10 Total predictions = 10 Prediction Accuracy = 100 %

FINAL LAB 1 RESULTS

Measure	Your Result
Total references in AI paper	10
References deeply audited	10
A: Verified	10
B: Wrong metadata	0

C: Frankenstein	0
D: Fabricated	0
E: Identifier mismatch	0
Authenticity Score	_100 /100
Prediction Accuracy	100 %

FINAL INTERPRETATION

1. Did professional-looking citations necessarily turn out to be genuine?

In my test run, all 10 citations ended up being real, but that isn't always a guarantee. Just because a reference has a realistic DOI, proper author names, and well-known journals doesn't automatically mean it exists. AI tools can easily generate smooth, authentic-looking citations that turn out to be fake upon closer inspection.

2. Did all genuine references actually support the claims for which they were cited?

Mostly, yes for basic concepts (like citing the original creators of SHAP and LIME).

3. What was the most serious citation-integrity problem you observed?

The biggest issue was claim exaggeration. Even though the paper exists and the details matched, the AI sometimes cited a study to support a specific comparison or point that wasn't actually tested or stated in that original paper.

4. What is the single most important lesson you learned from this lab?

Checking if a citation is real is only half the job. Just because a paper exists and the title matches doesn't mean the AI used it correctly. We always have to check the actual text to make sure it truly backs up what the AI claims.

SUBMISSION INSTRUCTIONS

Submit one PDF: Lab1_RollNumber_Name

Example: Lab1_IEC2026001_RahulSharma

STUDENT DECLARATION

I confirm that I generated the research paper using the recorded AI environment, preserved the original output before conducting the audit, followed the prescribed reference-selection procedure, and personally verified the evidence reported in this worksheet. I have not altered my pre-verification predictions after seeing the verification results.

Student Name: SRISHTI KUMARI

Roll Number: RSI2026514

Signature: srishtikumari

Date: 21/08/2026

CENTRAL LESSON OF LAB 1

Does the source exist? + Does the source actually support the claim?